SUSHMITHA SUDHARSAN

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SUMMARY

AI/ML Engineer with an MS in Data Analytics Engineering, specializing in production-grade MLOps and applied GenAI. Owned CI/CD for a 6-person MLOps platform on GCP with automated retraining and drift monitoring, serving under 200ms p99 and validating 2.5M+ records. Proficient in PyTorch, MLflow, Airflow, and Docker. First-author IEEE publication in applied deep learning.

EDUCATION

Northeastern University Apr 2026
Master of Science in Data Analytics Engineering, GPA: 3.92/4.0 *Boston, MA*
- **Coursework:** Data Mining, Generative AI, MLOps, Machine Learning, Deep Learning for AI, Data Visualization

SRM Institute of Science and Technology May 2024
Bachelor of Technology in Computer Science (AI & ML Specialization) *Chennai, India*

PROJECTS

Financial Stress Test Platform [GitHub] | *PyTorch, XGBoost, Airflow, MLflow, GCP* Sep – Dec 2025
- Built a platform with a 6-member team to stress-test how 100 companies would hold up if a past crisis (2008, COVID) happened again, automating PyTorch and XGBoost pipelines in Airflow and validating 2.5M+ records with Great Expectations
- Owned CI/CD for automated retraining, GCP deployment, and PyTest testing; tracked 50+ MLflow experiments and monitored drift weekly with Prometheus/Grafana at under 200ms p99

World Cup Analyst Chatbot [GitHub] | *LangChain, FAISS, GPT-4o, Gradio* Mar 2026
- Built a chatbot with a 3-member team that answers World Cup stats questions across 49,000+ matches and 47,000+ goals, using a ReAct agent that connects GPT-4o, embeddings, FAISS, and 5 tools
- Shipped in 48 hours with evidence-backed answers (confidence scores, stated limitations) and a match-outcome predictor

VisionScript: Multimodal Image Captioning [GitHub] | *CLIP, BLIP, GPT-2, PyTorch, Hugging Face* Jan – Apr 2026
- Designed the architecture with a 4-member team, connecting CLIP image embeddings to GPT-2 through a projection layer
- Built the BLIP app with single captions, paragraph captions, and visual Q&A, deployed as a live demo on Hugging Face Spaces

ICU Mortality & Length-of-Stay Prediction (MIMIC-IV) [GitHub] | *Python, Pandas, Scikit-learn, XGBoost* Jan – Apr 2025
- Built the length-of-stay regression models in a team of 2, using first-24-hour features from 94,000+ ICU admissions; all four regressors reached 0.94–0.96 days MAE, under the 1-day target
- Team's XGBoost mortality model reached 0.90 ROC-AUC; credentialed MIMIC-IV access under the PhysioNet data use agreement

EXPERIENCE

Data Analyst Intern Jun 2023 – Sep 2023
Whirl Data Inc. *Chennai, India*
- Analyzed 10,000+ dashcam frames to find 15+ factors (lighting, placement, plate condition, speed) hurting plate capture
- Built a Python/OpenCV pipeline scoring brightness, Laplacian sharpness, and contrast into 3 quality tiers; ˜10% unusable
- Extracted plate text with Tesseract OCR; proposed duplicate and anomaly-frame removal for plate-detection training

PUBLICATIONS & ACHIEVEMENTS

- **IEEE Publication (First Author), 2024 [IEEE Xplore]:** Developed an RNN/LSTM model to track classroom engagement and flag student fatigue, achieving 96% accuracy with 2–3 sec real-time latency across 30 participants
- **Data Mining Hackathon, 2nd place of 25 teams (0.89 F1):** Led a team building an ML pipeline in 48 hours

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, C++, Pandas, NumPy, Apache Airflow, ETL Pipelines, API Integration
Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM
GenAI & NLP: LLMs, RAG, LangChain, Multi-Agent Systems, NLP, Prompt Engineering
MLOps & Deployment: MLflow, Docker, CI/CD, FastAPI, Model Monitoring, Prometheus, Grafana, DVC, Great Expectations, PyTest
Cloud & Vision: GCP, Cloud Run, HuggingFace Spaces, YOLOv8, Object Detection, OpenCV, CLIP Embeddings

CERTIFICATIONS

- Azure Fundamentals (AZ-900) | Azure AI Fundamentals (AI-901) | Power BI | AWS Generative AI | AWS Prompt Engineering